$\forall\ h \in \mathbb{N}$, μ has no effect on Cov

$$\text{Cov}(\nabla S_t, \nabla S_{t-h}) = \text{Cov}(X_t, X_{t-h})$$

$$= \begin{cases} \sigma^2, & \text{if } h = 0 \\ 0, & \text{else} \end{cases}$$